



(L) Active and (R) passive 3-D glasses (courtesy: PVR Cinemas, Delhi)

The 'D' and 'E' of cinema

In digital 2-D cinema, there are two variants, D-cinema and E-cinema. For D-cinema, projectors and servers need to be DCI-compliant. To run a Hollywood movie, you have to satisfy this requirement. Also, currently for 3-D, you need the D-cinema set-up.

E-cinema is an initiative of the Indian film makers to release the Indian movies digitally. "For a drop of around 10 per cent in quality, we have huge price reduction in the E-cinema," explains Senthil. QubeXP-E is used for digital 2-D E-cinema. "The projectors use the DLP technology (not the DLP cinema technology) but do not adhere to DCI standards," adds Ganesh. These aren't 3-D-ready.

Microsoft's Windows XP—a modular embedded operating system—to maximise reliability and security. The server can be managed and operated locally or remotely via a powerful, highly interactive Web-based user interface. "We configure the channels to play either 2-D or 3-D content," explains Rajiv. Doremi is another popular vendor of playback servers in India.

3-D digital format

The fundamental working principle of 3-D is projecting left and right eye-view separately. As the 3-D movie carries the information of both images (left- and right-view), the 3-D file is almost double the size of 2-D file. Digital movie files come in DCI JPEG 2000 (J2K) file format created by Joint Photographic Experts Group. J2K supports

a very high resolution and flexibility of decoding algorithms and scalability. Earlier, MPEG file format was used and even today some of the films come in MPEG format.

"MPEG file for a 2-D version takes up to 80GB space, while J2K occupies up to 190-220 GB," shares Vardhan. Digital 3-D content is distributed in 48fps format.

3-D glasses

The perception of depth is possible only when you view the left- and right-eye images differently. All the 3-D technologies discussed so far work on the same principle. The passive polarisation technology used in dual projectors and colour wheel uses a passive glass filter. It uses polarised lenses for each eye and makes it possible to receive the images separately. RealD

is the market leader in these types of glasses. MasterImage 3-D too uses a similar technology.

You need to wear active glasses to view 3-D in XpanD technology. "The active glasses are called active because these have active electronics consisting of IR receiver and the battery," explains Ganesh. "The glass is a thin-film transistor liquid-crystal display (TFT LCD) that opens and shuts the LCD shutter for each eye in sync with the left and right projection so each eye sees the appropriate frame of the image."

Talking about the cost, Ganesh said that the passive glasses cost around 40-50 cents, whereas active glasses cost around \$50. Passive glasses are one-time-use, while active glasses can be cleaned and used again. "Till now, there is no technology that supports 3-D viewing without wearing glasses in theatres," says Ganesh. Many shopping malls, however, are showing 3-D without glasses, which has some sort of grids visible.

3-D cinema ecosystem

The 3-D ecosystem is the combination of technology and content. The technology has almost reached a sophisticated level, but the content is still in infancy. "Creating 3-D content is a major challenge," exclaims Senthil. "Just like the move from B&W to colour, a move is required for cinematographers, editors and directors to understand the rules of 3-D and what makes it interesting to the audience." Highly responsive image processing software are required to create 3-D content.

Animation and live-action content will hit the market first. "Some of the older classics like *Titanic* are already in process to be released in 3-D," claims Senthil. "We received a very good response for 3-D *Avatar* and we expect a lot more titles to be released in 3-D this year. Some Indian movies too are expected to hit the market in 3-D by the end of the year," informs Vardhan. ●

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